

Introduction to Interpretability and XAI

Our focus : Black box models

- Many current industrial challenges related to Uncertainty Quantification (UQ)
- How can we emulate these complex systems to convince the safety and regulatory bodies of their reliability?
- Design of a **surrogate model** that should be audited
- An example on **wind turbine**

Scientific context

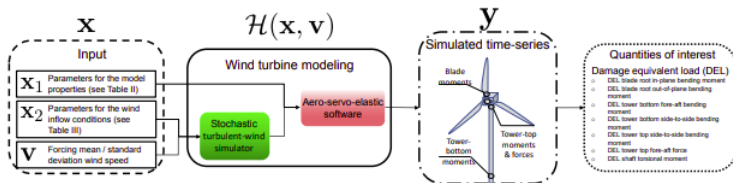


Figure 1. Wind turbine modeling framework, where $\mathbf{x}$ is the extended vector gathering both uncertainties from wind inflow parameters and model properties.

Extracted from Hirvoas et al. (2022)

Our focus : Black box models

- Our focus : methods that require nothing but input–output access to the model.
- No gradients, no differentiable architecture, no white-box access to weights, no retraining, no concept labels

Alternative approaches

- Identify zones of an image leading to a decision
- Popular methods : gradcam and extensions

See <https://www.irit.fr/Ronan.Sicre/enseignement/>



Illustration of a Gradcam explanation

Alternative approaches

- Identify parts of a neural networks involved in a decision
- Mechanistic interpretability and circuits

See

[https://www.neelnanda.io/mechanistic-interpretability/quicksta](https://www.neelnanda.io/mechanistic-interpretability/quickstart)

Alternative approaches

- Interpretate the latent space of a neural networks
- Concept bottleneck models

See <https://www.irit.fr/~Ronan.Sicre/enseignement/>

Three questions

- Which input factors drive outcomes across my whole operating range?
- Why did the model decide this for this one case?
- How can we turn these explanations into an actionable procedure?

The variable selection problem

- Aim : identify and eliminate, among for instance a hundred input variables influencing the phenomenon under study
- Distinguish those that have a negligible influence on the model's output from the other ones
- An answer : **variance-based sensitivity analysis and Shapley values**

The variable selection problem

- Aim : explain the input variables involved in output of a given instance
- An answer : **Shapley values**

Towards actionability

- Aim : beyond describing what mattered or why
- Identify what is to change, and gives you a valid guarantee about what will happen if you do
- Requirements
 - Mutability : the explanation must point to a variable an agent can actually manipulate
 - Causal validity : the explanation's implied effect of changing that variable must hold under intervention, not just under resampling/conditioning.
- An answer: **causally-aware sensitivity analysis**

Roadmap of the course

Schedule

- Lecture 1 : Variance based sensitivity analysis, Shapley values and friends
- Lecture 2 : Causality and causally aware importance measures